

ASSESSMENT TOOL – 1

Course Code	CSA0702	Course Title	Computer Networks
CO Assessed	CO5	Assessment	Constraint Problem Solving
Weightage		Total Marks	

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Section / Batch	ECE-2024	Date of Submission	19/06/2026
Faculty Incharge	Dr R S Selvan	Assessment	Individual Submission

Total Marks		Obtained Marks	
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Code of Conduct:

20/20


I Cindy R/192412571 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



note Rural Branch office

Problem Analysis

The branch office is located in a rural area where high-speed wired Internet is unavailable. It needs reliable connectivity for cloud applications, video conference, and secure communication with headquarters. Since the budget is limited, the solution should provide good performance without requiring expensive infrastructure.

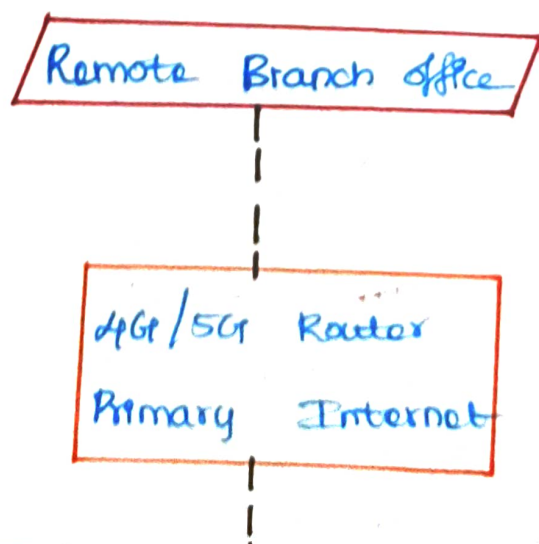
Recommended Networking Devices:

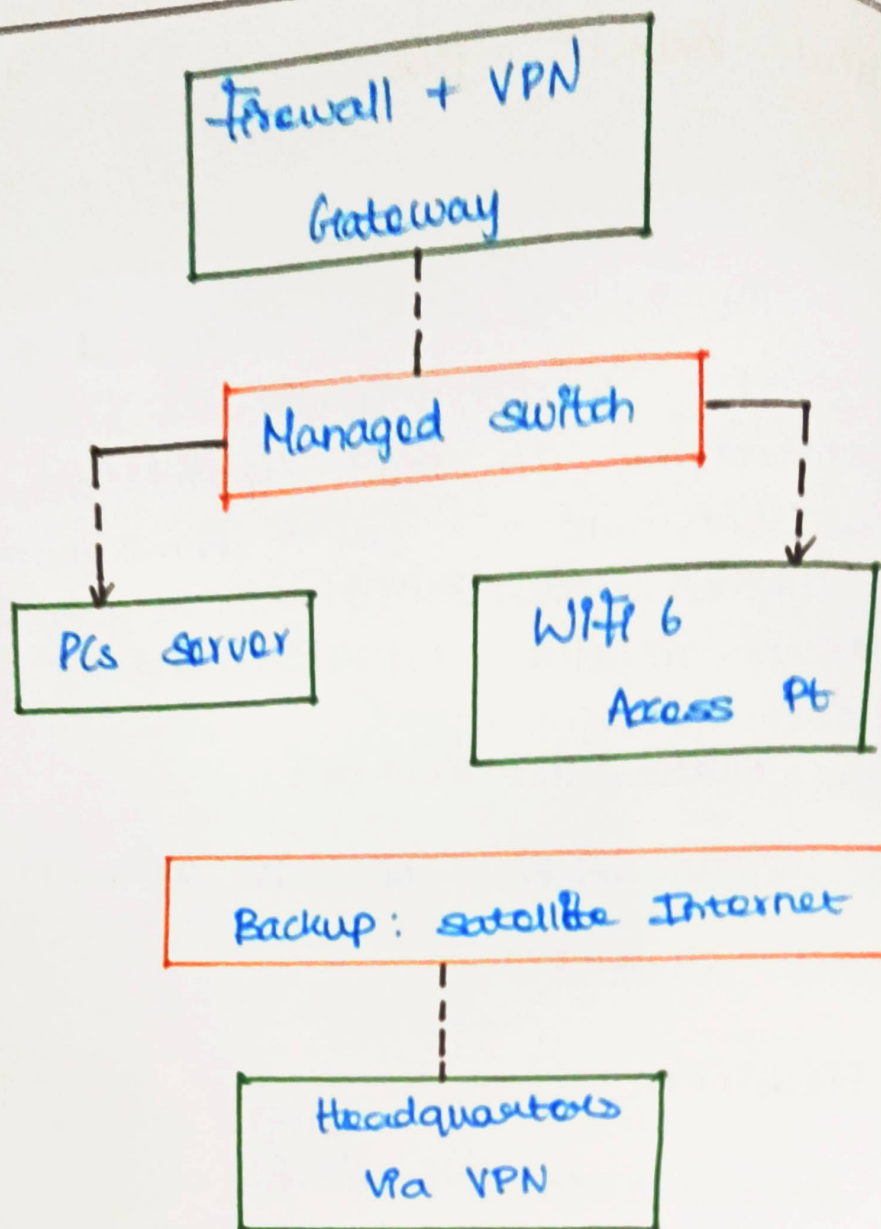
4G/5G Business router, Managed Ethernet switch, UPS for network equipment.

Security Mechanism:

- Firewall with intrusion prevention
- Regular security updates
- Site to site VPN
- WPA3

Block diagram:





Conclusion:

A 4G/5G connection with satellite backup, firewall, VPN and scalable LAN infrastructure provides a cost effective, secure and reliable solution for the rural branch.

2. University Wireless Network for 3000+ students

Problem Analysis:

The university requires campus-wide wireless connectivity for more than 3000 students across classrooms, laboratories, libraries and hostels. The major challenges are high user identity, security.

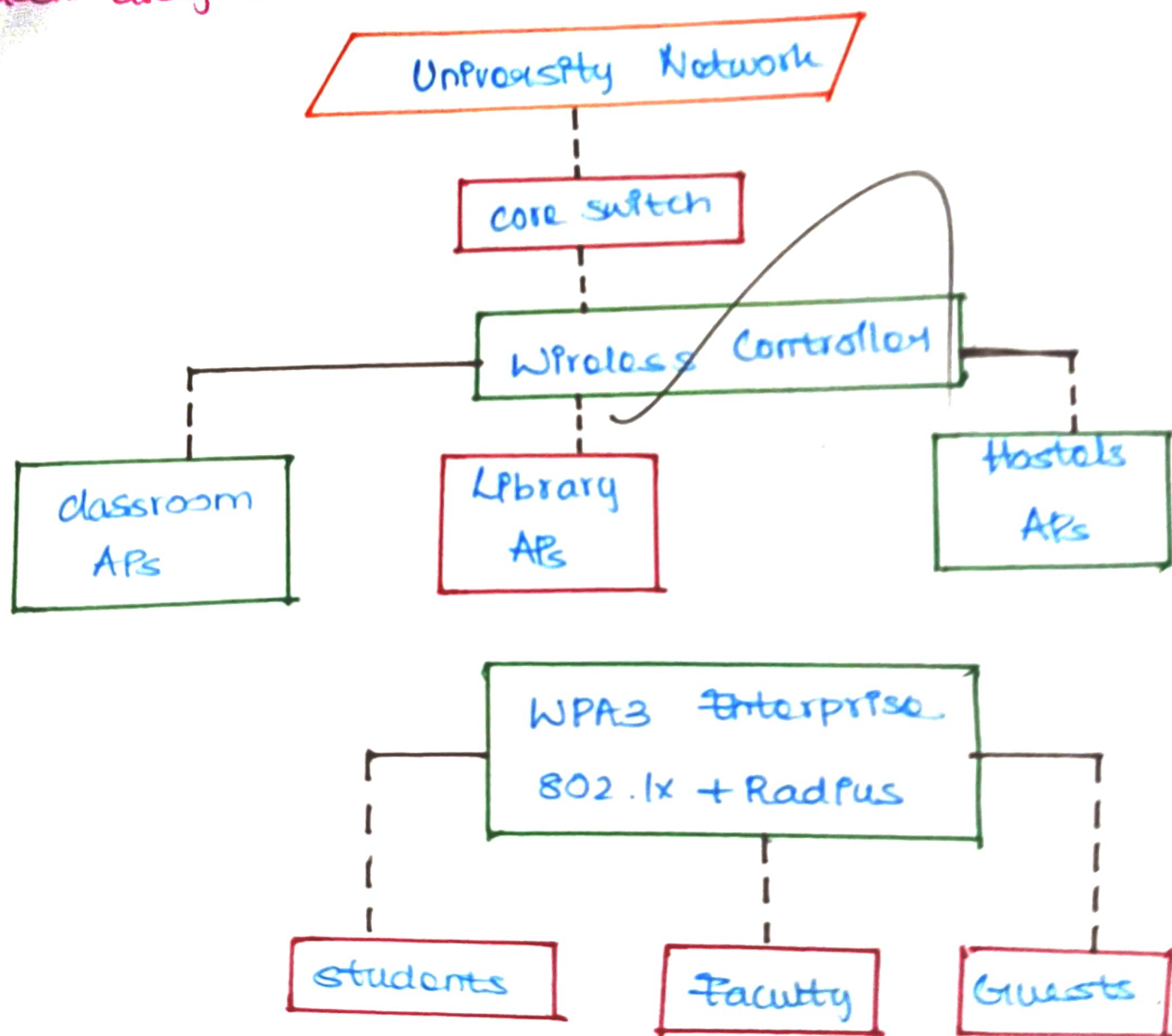
Recommended Network Design:

Use a centrally managed WPA3 6 network with strategically positioned access points. APs should be concentrated in high-density areas rather than distributed equally.

Authentication and security:

WPA3, 802.1x, RADIUS server, VLANs, Guest network with Internet-only access

Block diagram:



Conclusion:

A centrally managed WPA3 6 network with optimized AP placement, 5GHz as the primary band, and WPA3 - Enterprise authentication.

provides a good balance between coverage, cost, security, performance.

3. Security Network Architecture for a Financial Institution

Problem Analysis:

A financial institution handles sensitive customer and banking information, so strong security is required. However, security mechanisms should not significantly reduce network performance or increase operational costs.

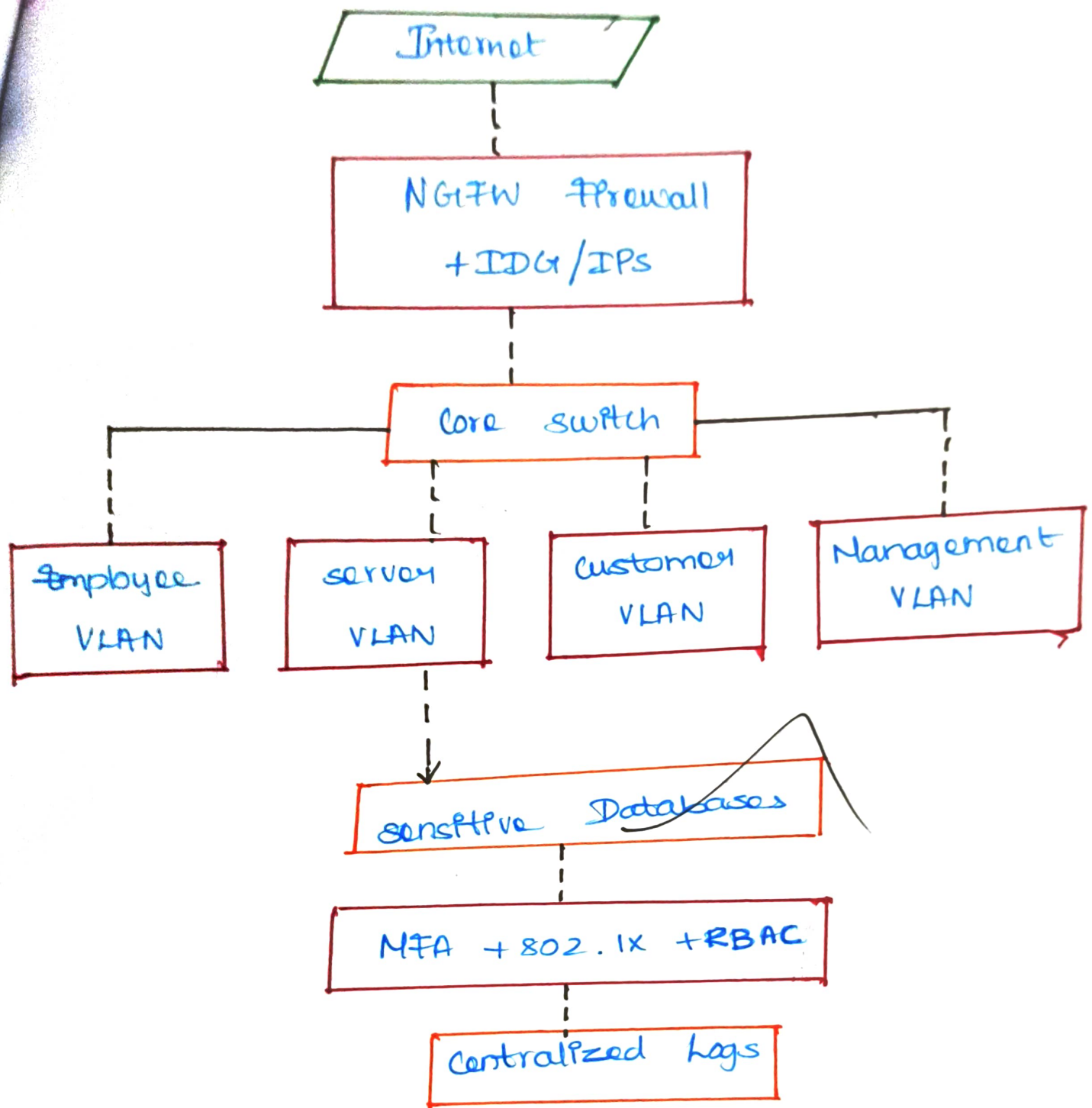
Network segmentation:

- * Employee VLAN
- * Server VLAN
- * Guest VLAN
- * Management VLAN
- * Customer - Service VLAN

Intrusion detection and Prevention:

- * Malware
- * Port scanning
- * Suspicious network behaviour
- * Known attack patterns
- * Unauthorized access attempts

Diagram:



Conclusion:

A segmented architecture using VLANs, NGFW, IDS/IPS provides strong cybersecurity while maintaining performance, controlling, supporting compliance, allowing future expansion.

ASD